

■ 시나리오 별 백업 복구(User Managed) - Archive log mode

KEY WORDS

- <시나리오 26> CFs 장애 복구(백업한 CFs 내용과 현재 DFs 정보가 달라졌을 경우)
- <시나리오 27> 모든 DFs, CFs 손상된 경우 복구
- <시나리오 28> System datafile, CFs 삭제된 경우 복구
- <시나리오 29> RFs, CFs 손상되었을 경우 복구
- <시나리오 30> 일반 DFs, inactive RF, CFs 손상된 경우 복구
- <시나리오 31> 일반 DFs, inactive RF, CFs 손상된 후에 log switch 가 발생했을 경우(hang 상태) 복구
- <시나리오 32> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 있을 경우)
- <시나리오 33> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 없을 경우), open backup 사용
- <시나리오 34> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 없을 경우), close backup 사용(DFs, CFs 만 복원)

<<시나리오 26>> CFs장애 복구(백업한 CFs 내용과 현재 DFs 정보가 달라졌을 경우)

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

--		
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
3321935	SYSTEM	
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
3321935	ONLINE	
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
3321935	ONLINE	
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf
3321935	ONLINE	

- 테이블스페이스 생성

```
SYS@orcl> create tablespace data1 datafile '/u01/app/oracle/oradata/ORCL/data01.dbf' size 5m;
Tablespace created.
```

- Alert log 확인

- 데이터파일이나 현재 구조가 변경되면 trace 생성.

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/backup/crecon_1.txt';
```

Database altered.

- Alert log 확인

```
SYS@orcl> create table hr.emp2 tablespace data1 as select * from hr.employees where department_id = 20;
```

Table created.

```
SYS@orcl> select count(*) from hr.emp2;
```

```
COUNT(*)
```

```
-----
```

```
2
```

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

ORA-00205: error in identifying control file, check alert log for more info

-- Alert log 확인

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl
```

ls: cannot access /u01/app/oracle/oradata/ORCL/control01.ctl: No such file or directory

■ Recovery

- 백업 컨트롤 파일 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/control01.ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
SYS@orcl> alter database mount;
```

Database altered.

- 컨트롤파일 trace

```
SYS@orcl> alter database backup controlfile to trace as '/home/oracle/backup/crecon2.txt';
```

Database altered.

- 데이터베이스 종료

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

* 데이터베이스를 비정상적인 종료했을 경우는 recover database 해야 한다.

```
SYS@orcl> !vi /home/oracle/backup/crecon2.txt
```

STARTUP NOMOUNT

CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS ARCHIVELOG

MAXLOGFILES 16

MAXLOGMEMBERS 3

MAXDATAFILES 100

MAXINSTANCES 8

MAXLOGHISTORY 292

LOGFILE

GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,

GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,

GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512

-- STANDBY LOGFILE

DATAFILE

'/u01/app/oracle/oradata/ORCL/system01.dbf',

'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',

'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',

'/u01/app/oracle/oradata/ORCL/users01.dbf'

CHARACTER SET AL32UTF8

;

RECOVER DATABASE

-- All logs need archiving and a log switch is needed.

```
ALTER SYSTEM ARCHIVE LOG ALL;
```

-- Database can now be opened normally.

ALTER DATABASE OPEN;

SYS@orcl> @/home/oracle/crecon2.txt

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 788529152 bytes

Database Buffers 335544320 bytes

Redo Buffers 7880704 bytes

Control file created.

ORA-00283: recovery session canceled due to errors

ORA-00264: no recovery required

System altered.

Database altered.

원래대로 돌리기

SYS@orcl> drop tablespace data1 including contents and datafiles;

Tablespace dropped.

SYS@orcl> shutdown immediate

Database closed.

Database dismounted.

ORACLE instance shut down.

SYS@orcl> startup

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

Database opened.

```
SSYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
3321911	1 /u01/app/oracle/oradata/ORCL/system01.dbf	3325696	NOT ACTIVE
3321911	3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf	3325696	NOT ACTIVE
3321911	4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf	3325696	NOT ACTIVE
3321911	7 /u01/app/oracle/oradata/ORCL/users01.dbf	3325696	NOT ACTIVE

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
3321935	1	13 /u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
3324481	2	14 /u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3321908	3	12 /u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
11	/home/oracle/arch1/arch_1_11_1201050570.arc	3321905	3321908
11	/home/oracle/arch2/arch_1_11_1201050570.arc	3321905	3321908
12	/home/oracle/arch1/arch_1_12_1201050570.arc	3321908	3321935
12	/home/oracle/arch2/arch_1_12_1201050570.arc	3321908	3321935
13	/home/oracle/arch1/arch_1_13_1201050570.arc	3321935	3324481

13 /home/oracle/arch2/arch_1_13_1201050570.arc

3321935

3324481

6 rows selected.

* online backup 진행.

<<시나리오 27>> 모든 DFs, CFs 손상된 경우 복구

```
SYS@orcl> alter system switch logfile;
System altered.
```

```
SYS@orcl> /
System altered.
```

```
SYS@orcl> /
System altered.
```

```
SYS@orcl> /
System altered.
```

```
SYS@orcl> /
System altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
3326481	3326484			YES
2	20	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3326484	1.8447E+19			NO
3	18	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
3326478	3326481			YES

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
CHANGE#	TIME		
1	/u01/app/oracle/oradata/ORCL/system01.dbf	3326484	NOT ACTIVE
3326435	2025-05-14 21:44:23.78263		

```

3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf 3326484 NOT ACTIVE
3326435 2025-05-14 21:44:23.78263
4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf 3326484 NOT ACTIVE
3326435 2025-05-14 21:44:23.78263
7 /u01/app/oracle/oradata/ORCL/users01.dbf 3326484 NOT ACTIVE
3326435 2025-05-14 21:44:23.78263

```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
11	/home/oracle/arch1/arch_1_11_1201050570.arc	3321905	3321908
11	/home/oracle/arch2/arch_1_11_1201050570.arc	3321905	3321908
12	/home/oracle/arch1/arch_1_12_1201050570.arc	3321908	3321935
12	/home/oracle/arch2/arch_1_12_1201050570.arc	3321908	3321935
13	/home/oracle/arch1/arch_1_13_1201050570.arc	3321935	3324481
13	/home/oracle/arch2/arch_1_13_1201050570.arc	3321935	3324481
14	/home/oracle/arch1/arch_1_14_1201050570.arc	3324481	3326459
14	/home/oracle/arch2/arch_1_14_1201050570.arc	3324481	3326459
15	/home/oracle/arch1/arch_1_15_1201050570.arc	3326459	3326472
15	/home/oracle/arch2/arch_1_15_1201050570.arc	3326459	3326472
16	/home/oracle/arch1/arch_1_16_1201050570.arc	3326472	3326475

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
16	/home/oracle/arch2/arch_1_16_1201050570.arc	3326472	3326475
17	/home/oracle/arch1/arch_1_17_1201050570.arc	3326475	3326478
17	/home/oracle/arch2/arch_1_17_1201050570.arc	3326475	3326478
18	/home/oracle/arch1/arch_1_18_1201050570.arc	3326478	3326481
18	/home/oracle/arch2/arch_1_18_1201050570.arc	3326478	3326481
19	/home/oracle/arch1/arch_1_19_1201050570.arc	3326481	3326484
19	/home/oracle/arch2/arch_1_19_1201050570.arc	3326481	3326484

```
18 rows selected.
```

- 업무 중

```
SYS@orcl> create table hr.emp3 tablespace users as select * from hr.employees where department_id = 50;
Table created.
```

```
SYS@orcl> select count(*) from hr.emp3;
```


COUNT(*)

45

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,dbf
/u01/app/oracle/oradata/ORCL/control01.ctl  /u01/app/oracle/oradata/ORCL/system01.dbf
/u01/app/oracle/oradata/ORCL/control02.ctl  /u01/app/oracle/oradata/ORCL/temp01.dbf
/u01/app/oracle/oradata/ORCL/data01.dbf      /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/sysaux01.dbf    /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/*.ctl: No such file or directory
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
/u01/app/oracle/oradata/orcl/*.dbf: No such file or directory
```

새로운 세션

```
[oracle@oel7v9r1 ~]$ sqlplus hr/hr
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed May 14 21:49:46 2025
Version 19.3.0.0.0
```

Copyright (c) 1982, 2019, Oracle. All rights reserved.

ERROR:

```
ORA-00604: error occurred at recursive SQL level 1
ORA-01116: error in opening database file 1
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

-- Alert log 확인

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
initorcl.ora redo01.log redo02.log redo03.log
```

■ Recovery

- 데이터베이스 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

1. 데이터파일, 컨트롤파일, 백업본 찾아 restore

```
SYS@orcl> ! ls /home/oracle/backup/open_bkp/*.{dbf,ctl}
/home/oracle/backup/open_bkp/control01.ctl          /home/oracle/backup/open_bkp/system01.dbf
/home/oracle/backup/open_bkp/users01.dbf
/home/oracle/backup/open_bkp/data01.dbf            /home/oracle/backup/open_bkp/temp01.dbf
/home/oracle/backup/open_bkp/sysaux01.dbf          /home/oracle/backup/open_bkp/undotbs01.dbf
```

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers            620756992 bytes
Redo Buffers                 7876608 bytes
Database mounted.
```

2. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover database using backup controlfile
ORA-00279: change 3326435 generated at 05/14/2025 21:44:23 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_14_1201050570.arc
ORA-00280: change 3326435 for thread 1 is in sequence #14
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
```

```
ORA-00279: change 3326459 generated at 05/14/2025 21:44:27 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_15_1201050570.arc
ORA-00280: change 3326459 for thread 1 is in sequence #15
ORA-00278: log file '/home/oracle/arch2/arch_1_14_1201050570.arc' no longer needed for this recovery
```

ORA-00279: change 3326472 generated at 05/14/2025 21:44:51 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_16_1201050570.arc
ORA-00280: change 3326472 for thread 1 is in sequence #16
ORA-00278: log file '/home/oracle/arch2/arch_1_15_1201050570.arc' no longer needed for this recovery

ORA-00279: change 3326475 generated at 05/14/2025 21:44:52 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_17_1201050570.arc
ORA-00280: change 3326475 for thread 1 is in sequence #17
ORA-00278: log file '/home/oracle/arch2/arch_1_16_1201050570.arc' no longer needed for this recovery

ORA-00279: change 3326478 generated at 05/14/2025 21:44:53 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_18_1201050570.arc
ORA-00280: change 3326478 for thread 1 is in sequence #18
ORA-00278: log file '/home/oracle/arch2/arch_1_17_1201050570.arc' no longer needed for this recovery

ORA-00279: change 3326481 generated at 05/14/2025 21:44:56 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_19_1201050570.arc
ORA-00280: change 3326481 for thread 1 is in sequence #19
ORA-00278: log file '/home/oracle/arch2/arch_1_18_1201050570.arc' no longer needed for this recovery

ORA-00279: change 3326484 generated at 05/14/2025 21:44:56 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_20_1201050570.arc
ORA-00280: change 3326484 for thread 1 is in sequence #20
ORA-00278: log file '/home/oracle/arch2/arch_1_19_1201050570.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_20_1201050570.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

SYS@orcl> recover database using backup controlfile
ORA-00279: change 3326484 generated at 05/14/2025 21:44:56 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_20_1201050570.arc

ORA-00280: change 3326484 for thread 1 is in sequence #20

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

/u01/app/oracle/oradata/ORCL/redo02.log

Log applied.

Media recovery complete.

-- Alert log

Thread 1 advanced to log sequence 20 (LGWR switch)

Current log# 2 seq# 20 mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

3. 데이터베이스 resetlogs open

SYS@orcl> alter database open resetlogs;

Database altered.

SYS@orcl> select status from v\$instance;

STATUS

OPEN

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	
ARC	FIRST_CHANGE#	NEXT_CHANGE#			

1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT	NO
3327680	1.8447E+19				
2	0	/u01/app/oracle/oradata/ORCL/redo02.log	10	UNUSED	YES
0	0				
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED	YES
0	0				

SYS@orcl> select count(*) from hr.emp3;

COUNT(*)

45

* online backup 진행.

<<시나리오 28>> System datafile, CFs 삭제된 경우 복구

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
-- 다른세션
```

```
[oracle@oel7v9r1 ~]$ sqlplus hr/hr
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed May 14 22:02:22 2025  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
ERROR:
```

```
ORA-00604: error occurred at recursive SQL level 1
```

```
ORA-01116: error in opening database file 1
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 3
```

```
ORA-00604: error occurred at recursive SQL level 1
```

```
ORA-01116: error in opening database file 1
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 3
```

```
-- Alert log 확인
```

```
-- sys
```

```
SYS@orcl> select * from v$database;
```

```
select * from v$database
```

```
*
```

```
ERROR at line 1:
```

```
ORA-00210: cannot open the specified control file
```

```
ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
OPEN
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
```

```
data01.dbf  initorcl.ora  redo01.log  redo02.log  redo03.log  sysaux01.dbf  temp01.dbf      undotbs01.dbf
users01.dbf
```

■ Recovery

- 데이터베이스 종료

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

1. 시스템 데이터파일, 컨트롤 파일 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/system01.dbf /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/control01.ctl /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount
```

```
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size              905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
Database mounted.
```

3. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover database using backup controlfile
```

```
ORA-00279: change 3332042 generated at 05/14/2025 22:01:05 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201125487.arc
```

```
ORA-00280: change 3332042 for thread 1 is in sequence #1
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00279: change 3332068 generated at 05/14/2025 22:01:08 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201125487.arc
```

```
ORA-00280: change 3332068 for thread 1 is in sequence #2
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201125487.arc' no longer needed for this recovery
```

```
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_2_1201125487.arc'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 7
```

```
SYS@orcl> recover database using backup controlfile
```

```
ORA-00279: change 3332068 generated at 05/14/2025 22:01:08 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201125487.arc
```

```
ORA-00280: change 3332068 for thread 1 is in sequence #2
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
/u01/app/oracle/oradata/ORCL/redo02.log
```

```
Log applied.
```

```
Media recovery complete.
```

4. resetlogs로 open

```
SYS@orcl> alter database open resetlogs;
```

```
Database altered.
```

```
# 정보 확인
```



```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
OPEN
```

* online backup 진행.

<<시나리오 29>> RFs, CFs 손상되었을 경우 복구

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,log}
/u01/app/oracle/oradata/ORCL/control01.ctl /u01/app/oracle/oradata/ORCL/redo02.log
/u01/app/oracle/oradata/ORCL/redo01.log /u01/app/oracle/oradata/ORCL/redo03.log
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,log}
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,log}
ls: cannot access /u01/app/oracle/oradata/ORCL/*.ctl: No such file or directory
ls: cannot access /u01/app/oracle/oradata/ORCL/*.log: No such file or directory
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

-- Alert log 확인

```
SYS@orcl> select status from v$instance;
select status from v$instance
*
ERROR at line 1:
```

안되면 shut abort

■ Recovery

1. 모든 데이터파일, 컨트롤파일 백업 본 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.dbf,ctl} /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/data01.dbf' -> '/u01/app/oracle/oradata/ORCL/data01.dbf'
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount
```

```
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size              905969664 bytes
Database Buffers          620756992 bytes
Redo Buffers               7876608 bytes
Database mounted.
```

3. 백업 이후 변경정보(리두) 적용 cancel base recovery

```
SYS@orcl> recover database using backup controlfile until cancel
```

```
ORA-00279: change 3332915 generated at 05/14/2025 22:11:09 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201126180.arc
```

```
ORA-00280: change 3332915 for thread 1 is in sequence #1
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00279: change 3332939 generated at 05/14/2025 22:11:13 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201126180.arc
```

```
ORA-00280: change 3332939 for thread 1 is in sequence #2
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201126180.arc' no longer needed for this recovery
```

```
ORA-00279: change 3332946 generated at 05/14/2025 22:11:28 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201126180.arc
```

```
ORA-00280: change 3332946 for thread 1 is in sequence #3
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_2_1201126180.arc' no longer needed for this recovery
```

```
ORA-00279: change 3332949 generated at 05/14/2025 22:11:33 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_4_1201126180.arc
```

```
ORA-00280: change 3332949 for thread 1 is in sequence #4
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_3_1201126180.arc' no longer needed for this recovery
```

```
ORA-00279: change 3332952 generated at 05/14/2025 22:11:33 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_5_1201126180.arc
```

```
ORA-00280: change 3332952 for thread 1 is in sequence #5
```

```
ORA-00278: log file '/home/oracle/arch2/arch_1_4_1201126180.arc' no longer needed for this recovery
```

ORA-00279: change 3332956 generated at 05/14/2025 22:11:36 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_6_1201126180.arc
ORA-00280: change 3332956 for thread 1 is in sequence #6
ORA-00278: log file '/home/oracle/arch2/arch_1_5_1201126180.arc' no longer needed for this recovery

ORA-00279: change 3332959 generated at 05/14/2025 22:11:38 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201126180.arc
ORA-00280: change 3332959 for thread 1 is in sequence #7
ORA-00278: log file '/home/oracle/arch2/arch_1_6_1201126180.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_7_1201126180.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3332959 generated at 05/14/2025 22:11:38 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201126180.arc
ORA-00280: change 3332959 for thread 1 is in sequence #7

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
Media recovery cancelled.

4. 데이터베이스 open resetlogs

SYS@orcl> alter database open resetlogs;
Database altered.

* online backup 진행.

<<시나리오 30>> 일반 DFs, inactive RF, CFs 손상된 경우 복구

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo01.log
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
```

```
data01.dbf  initorcl.ora  redo02.log  redo03.log  sysaux01.dbf      system01.dbf  temp01.dbf  undotbs01.dbf
```

```
SYS@orcl> shutdown immediate
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 11267
```

```
Session ID: 1 Serial number: 37031
```

```
-- Alert log 확인
```

```
SYS@orcl> conn / as sysdba
```

```
Connected to an idle instance.
```

```
SSYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
```

```
data01.dbf  initorcl.ora  redo02.log  redo03.log  sysaux01.dbf      system01.dbf  temp01.dbf  undotbs01.dbf
```

■ Recovery

1. 문제가 발생한 데이터파일, 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.ctl /u01/app/oracle/oradata/ORCL/
```

```
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
'/home/oracle/backup/open_bkp/data01.dbf' -> '/u01/app/oracle/oradata/ORCL/data01.dbf'
```

```
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
```

```
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
```

```
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
```

```
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size              905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
Database mounted.
```

3. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover database using backup controlfile
ORA-00279: change 3333771 generated at 05/14/2025 22:29:22 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201127332.arc
ORA-00280: change 3333771 for thread 1 is in sequence #1
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
```

```
ORA-00279: change 3333795 generated at 05/14/2025 22:29:25 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201127332.arc
ORA-00280: change 3333795 for thread 1 is in sequence #2
ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201127332.arc' no longer needed for this recovery
```

```
ORA-00279: change 3333804 generated at 05/14/2025 22:29:44 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201127332.arc
ORA-00280: change 3333804 for thread 1 is in sequence #3
ORA-00278: log file '/home/oracle/arch2/arch_1_2_1201127332.arc' no longer needed for this recovery
```

```
ORA-00279: change 3333807 generated at 05/14/2025 22:29:46 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_4_1201127332.arc
ORA-00280: change 3333807 for thread 1 is in sequence #4
ORA-00278: log file '/home/oracle/arch2/arch_1_3_1201127332.arc' no longer needed for this recovery
```

```
ORA-00279: change 3333810 generated at 05/14/2025 22:29:46 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_5_1201127332.arc
```

ORA-00280: change 3333810 for thread 1 is in sequence #5

ORA-00278: log file '/home/oracle/arch2/arch_1_4_1201127332.arc' no longer needed for this recovery

ORA-00279: change 3333813 generated at 05/14/2025 22:29:49 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_6_1201127332.arc

ORA-00280: change 3333813 for thread 1 is in sequence #6

ORA-00278: log file '/home/oracle/arch2/arch_1_5_1201127332.arc' no longer needed for this recovery

ORA-00279: change 3333816 generated at 05/14/2025 22:29:50 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201127332.arc

ORA-00280: change 3333816 for thread 1 is in sequence #7

ORA-00278: log file '/home/oracle/arch2/arch_1_6_1201127332.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_7_1201127332.arc'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> recover database using backup controlfile

ORA-00279: change 3333816 generated at 05/14/2025 22:29:50 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201127332.arc

ORA-00280: change 3333816 for thread 1 is in sequence #7

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

Media recovery cancelled.

4. 데이터베이스 resetlogs open

SYS@orcl> alter database open resetlogs;

Database altered.

* online backup 진행.

<<시나리오 31>> 일반 DFs, inactive RF, CFs 손상된 후에 log switch가 발생했을 경우(hang상태) 복구

■ 장애 발생

```
SYS@orcl> alter system switch logfile;(여러번)
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo01.log
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
```

```
data01.dbf  initorcl.ora  redo02.log  redo03.log  sysaux01.dbf      system01.dbf  temp01.dbf  undotbs01.dbf
```

```
SYS@orcl> alter system switch logfile;
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

```
/ <-- 행 상태 발견
```

```
SYS@orcl> select * from v$database;((?))
```

```
SYS@orcl> shutdown abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
```

```
data01.dbf  initorcl.ora  redo02.log  redo03.log  sysaux01.dbf      system01.dbf  temp01.dbf  undotbs01.dbf
```

■ Recovery

1. 모든 데이터파일, 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.{dbf,ctl} /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/data01.dbf' -> '/u01/app/oracle/oradata/ORCL/data01.dbf'
```



```
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount  
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes  
Fixed Size                  8896880 bytes  
Variable Size              905969664 bytes  
Database Buffers          620756992 bytes  
Redo Buffers               7876608 bytes  
Database mounted.
```

3. 백업 이후 변경정보(리두) 적용, cancel base recovery

```
SYS@orcl> recover database using backup controlfile until cancel  
ORA-00279: change 3334565 generated at 05/14/2025 22:36:37 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201127773.arc  
ORA-00280: change 3334565 for thread 1 is in sequence #1
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
auto
```

```
ORA-00279: change 3334590 generated at 05/14/2025 22:36:40 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201127773.arc  
ORA-00280: change 3334590 for thread 1 is in sequence #2  
ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201127773.arc' no longer needed for this recovery
```

```
ORA-00279: change 3334603 generated at 05/14/2025 22:37:16 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201127773.arc  
ORA-00280: change 3334603 for thread 1 is in sequence #3  
ORA-00278: log file '/home/oracle/arch2/arch_1_2_1201127773.arc' no longer needed for this recovery
```

ORA-00279: change 3334607 generated at 05/14/2025 22:37:19 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_4_1201127773.arc
ORA-00280: change 3334607 for thread 1 is in sequence #4
ORA-00278: log file '/home/oracle/arch2/arch_1_3_1201127773.arc' no longer needed for this recovery

ORA-00279: change 3334610 generated at 05/14/2025 22:37:21 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_5_1201127773.arc
ORA-00280: change 3334610 for thread 1 is in sequence #5
ORA-00278: log file '/home/oracle/arch2/arch_1_4_1201127773.arc' no longer needed for this recovery

ORA-00279: change 3334613 generated at 05/14/2025 22:37:22 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_6_1201127773.arc
ORA-00280: change 3334613 for thread 1 is in sequence #6
ORA-00278: log file '/home/oracle/arch2/arch_1_5_1201127773.arc' no longer needed for this recovery

ORA-00279: change 3334616 generated at 05/14/2025 22:37:23 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201127773.arc
ORA-00280: change 3334616 for thread 1 is in sequence #7
ORA-00278: log file '/home/oracle/arch2/arch_1_6_1201127773.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_7_1201127773.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7

SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3334616 generated at 05/14/2025 22:37:23 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_7_1201127773.arc
ORA-00280: change 3334616 for thread 1 is in sequence #7

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
Media recovery cancelled.

4. 데이터베이스 open resetlogs

SYS@oracl> alter database open resetlogs;

Database altered.

* online backup 진행.

<<시나리오 32>> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 있을 경우)

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf,log}
```

-- Alert log 확인

```
SYS@orcl> shutdown abort
```

ORACLE instance shut down.

■ Recovery

1. 모든 데이터파일, 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.ctl,dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'  
'/home/oracle/backup/open_bkp/data01.dbf' -> '/u01/app/oracle/oradata/ORCL/data01.dbf'  
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount
```

ORACLE instance started.

```
Total System Global Area 1543500144 bytes  
Fixed Size                  8896880 bytes  
Variable Size              905969664 bytes  
Database Buffers          620756992 bytes  
Redo Buffers               7876608 bytes  
Database mounted.
```

3. 백업이후 변경정보(리두) 적용, cancel base recovery

```
SYS@orcl> recover database using backup controlfile until cancel
```

ORA-00279: change 3336678 generated at 05/14/2025 23:02:53 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_1_1201128295.arc

ORA-00280: change 3336678 for thread 1 is in sequence #1

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

auto

ORA-00279: change 3336702 generated at 05/14/2025 23:02:56 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201128295.arc

ORA-00280: change 3336702 for thread 1 is in sequence #2

ORA-00278: log file '/home/oracle/arch2/arch_1_1_1201128295.arc' no longer needed for this recovery

ORA-00279: change 3336711 generated at 05/14/2025 23:03:09 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_3_1201128295.arc

ORA-00280: change 3336711 for thread 1 is in sequence #3

ORA-00278: log file '/home/oracle/arch2/arch_1_2_1201128295.arc' no longer needed for this recovery

ORA-00279: change 3336742 generated at 05/14/2025 23:04:25 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_4_1201128295.arc

ORA-00280: change 3336742 for thread 1 is in sequence #4

ORA-00278: log file '/home/oracle/arch2/arch_1_3_1201128295.arc' no longer needed for this recovery

ORA-00279: change 3336745 generated at 05/14/2025 23:04:26 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_5_1201128295.arc

ORA-00280: change 3336745 for thread 1 is in sequence #5

ORA-00278: log file '/home/oracle/arch2/arch_1_4_1201128295.arc' no longer needed for this recovery

ORA-00279: change 3336748 generated at 05/14/2025 23:04:28 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_6_1201128295.arc

ORA-00280: change 3336748 for thread 1 is in sequence #6

ORA-00278: log file '/home/oracle/arch2/arch_1_5_1201128295.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_6_1201128295.arc'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3336748 generated at 05/14/2025 23:04:28 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_6_1201128295.arc
ORA-00280: change 3336748 for thread 1 is in sequence #6
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
Media recovery cancelled.
```

4. 데이터베이스 open resetlogs

```
SYS@orcl> alter database open resetlogs;
```

Database altered.

* online backup 진행.

<<시나리오 33>> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 없을 경우), open backup사용

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf,log}
```

```
SYS@orcl> ! rm /home/oracle/arch{1,2}/*
```

```
SYS@orcl> shutdown abort  
ORACLE instance shut down.
```

■ Recovery

1. 모든 데이터파일, 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.ctl,dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'  
'/home/oracle/backup/open_bkp/data01.dbf' -> '/u01/app/oracle/oradata/ORCL/data01.dbf'  
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount  
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes  
Fixed Size                  8896880 bytes  
Variable Size               905969664 bytes  
Database Buffers           620756992 bytes  
Redo Buffers                 7876608 bytes  
Database mounted.
```

3. 백업이후 변경정보(리두) 적용, cancel base recovery

```
SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3337546 generated at 05/14/2025 23:09:09 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201129693.arc
ORA-00280: change 3337546 for thread 1 is in sequence #2
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_2_1201129693.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
```

```
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_2_1201129693.arc'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
```

```
ORA-01547: warning: RECOVER succeeded but OPEN RESETLOGS would get error below
ORA-01195: online backup of file 1 needs more recovery to be consistent
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3337546 generated at 05/14/2025 23:09:09 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_2_1201129693.arc
ORA-00280: change 3337546 for thread 1 is in sequence #2
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
ORA-01547: warning: RECOVER succeeded but OPEN RESETLOGS would get error below
ORA-01195: online backup of file 1 needs more recovery to be consistent
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-01112: media recovery not started
```


4. 데이터베이스 강제 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

5. pfile에 _allow_resetlogs_corruption=true 추가한 뒤 pfile로 데이터베이스 mount

```
SYS@orcl> create pfile from spfile;
File created.
```

```
SYS@orcl> !vi $ORACLE_HOME/dbs/initiorcl.ora
```

_allow_resetlogs_corruption=true <-- 추가

```
SYS@orcl> startup pfile=$ORACLE_HOME/dbs/initiorcl.ora mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers            620756992 bytes
Redo Buffers                 7876608 bytes
Database mounted.
```

6. 데이터베이스 open resetlogs

```
SYS@orcl> alter database open resetlogs;
```

Database altered.

* online backup 진행.

<<시나리오34>> 모든 DFs, CFs, RFs 삭제된 경우 복구(단 백업 이후에 아카이브 없을 경우),
close backup사용(DFs, CFs만 복원)

■ 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf,log}
```

```
SYS@orcl> ! rm /home/oracle/arch{1,2}/*
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> /
```

System altered.

```
SYS@orcl> /
```

```
^Calter system switch logfile
```

```
*
```

ERROR at line 1:

ORA-01013: user requested cancel of current operation

```
SYS@orcl> shutdown abort
```

■ Recovery

1. 모든 데이터파일, 컨트롤파일 백업본을 찾아 restore

```
SYS@orcl> !cp -v /home/oracle/backup/close_bkp/*.dbf,ctl /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/close_bkp/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'  
'/home/oracle/backup/close_bkp/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'  
'/home/oracle/backup/close_bkp/initiorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initiorcl.ora'  
'/home/oracle/backup/close_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'/home/oracle/backup/close_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'/home/oracle/backup/close_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'/home/oracle/backup/close_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'/home/oracle/backup/close_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'  
.dbf'
```

2. 데이터베이스 mount

```
SYS@orcl> startup mount
```

```
SYS@orcl> recover database using backup controlfile until cancel
```

```
ORA-00279: change 3105439 generated at 05/13/2025 22:30:41 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_8_1201040433.arc
```

```
ORA-00280: change 3105439 for thread 1 is in sequence #8
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_8_1201040433.arc'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 7
```

```
ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_8_1201040433.arc'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 7
```

```
SYS@orcl> recover database using backup controlfile until cancel
```

```
ORA-00279: change 3105439 generated at 05/13/2025 22:30:41 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_8_1201040433.arc
```

```
ORA-00280: change 3105439 for thread 1 is in sequence #8
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
cancel
```

```
Media recovery cancelled.
```

```
SYS@orcl> alter database open resetlogs;
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER
--------	-----------	--------

MB	STATUS
----	--------

ARC FIRST_CHANGE# NEXT_CHANGE#

1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE	YES
3105440	3108676				
2	2	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT	NO
3108676	1.8447E+19				
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED	YES
0	0				

SYS@orcl> select status from v\$instance;

STATUS

OPEN

* online backup 진행.